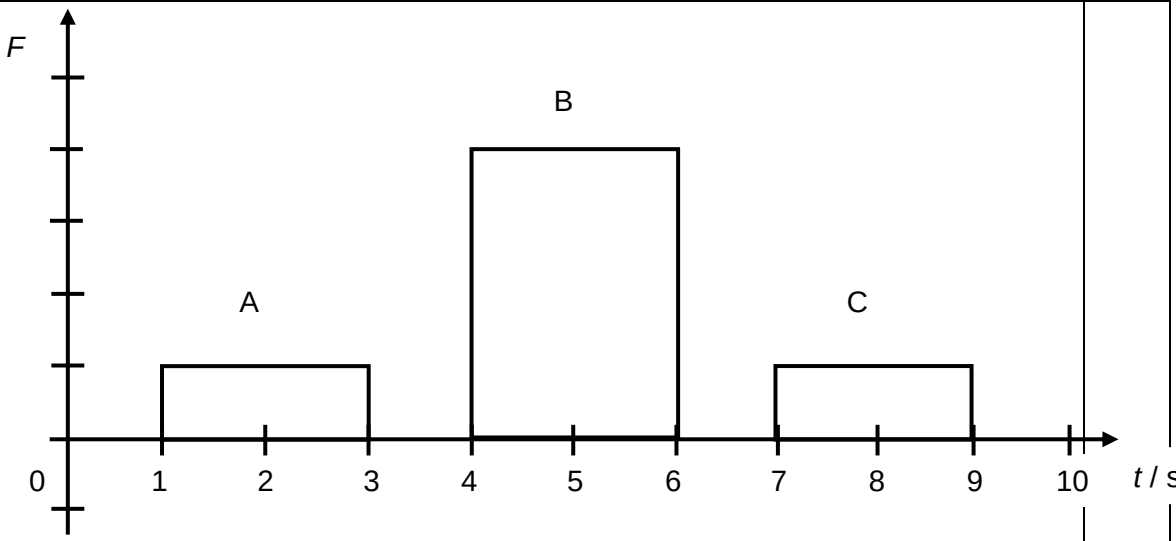


	By <u>Newton's Third Law</u> , the force on the magnet is <u>into the plane</u> of paper.	B1
8aii	Force <u>decreases</u> by <u>half / to half of its original value</u> . OR from a <u>maximum value at $\theta = 0^\circ$ to a half its maximum value at $\theta = 60^\circ$</u> .	M1 A1
8bi	Magnetic force = Bqv $= 0.45 (1.60 \times 10^{-19})(5.0)$ $= 3.6 \times 10^{-19} \text{ N}$	M1 A0
8bii	By Fleming's Right Hand Rule, there is <u>rate of cutting of flux</u> which <u>induced current to flow from B to A</u> through the wire. OR By Fleming's Left Hand Rule, <u>magnetic force</u> acting on electron <u>is directed towards B</u> . Electrons accumulate at end B leaving <u>excess positive charges at end A</u> . End <u>A</u> has a higher potential.	M1 A1
8biii1	Potential difference across the two ends of wire <u>produces an electric field</u> . Electrons in the wire experience an <u>electric force of equal magnitude but directed oppositely to the magnetic force</u> . Hence, equilibrium is achieved.	B1 B1
8biii2	Electron in equilibrium, $F_B = Bqv = qE$, so $E = F_B / q$ $E = 3.6 \times 10^{-19} / 1.60 \times 10^{-19}$ $= 2.25 = 2.3 \text{ N C}^{-1}$	C1 A1
8biii3	$E = \Delta V / d$ don't accept using e.m.f. = BLv $2.25 = \Delta V / 0.20$ $\Delta V = 0.45 \text{ V}$	A1
8ci	The frame experiences <u>an increase in flux linkage</u> . By Faraday's law, an <u>emf is induced</u> across XY. By Lenz's Law, <u>a current is induced</u> in the frame and flows clockwise ($X \rightarrow Y \rightarrow Z \rightarrow W$), resulting in a <u>magnetic force on XY to the left / against its motion</u> . To maintain constant speed, there should be <u>no net force</u> . Hence, an external force needs to be applied to the <u>right</u> . OR	M1 M1 A1 M1

	The frame experiences <u>an increase in flux linkage</u>. By Faraday's law, an <u>emf is induced</u> across XY. A <u>current is induced</u> in the frame, <u>if no external force</u> is applied, <u>kinetic energy of the frame will be transformed to thermal energy</u> To maintain constant speed, <u>work has to be done by an external force</u> so that the <u>kinetic energy is maintained</u>.	M1 A1
8cii	Induced emf = BLv $= (1.0)(0.020)(0.010)$ $= 2.0 \times 10^{-4} \text{ V}$ $F = BIL$ $= (1.0) \left[\frac{(2.0 \times 10^{-4})}{8.0 \times 10^{-4}} \right] (0.020)$ $= 5.0 \times 10^{-3} \text{ N}$ OR Induced emf = BLv $= (1.0)(0.020)(0.010)$ $= 2.0 \times 10^{-4} \text{ V}$ Heat dissipated per unit time = $\frac{V^2}{R}$ $= \frac{(2.0 \times 10^{-4})^2}{8.0 \times 10^{-4}}$ $= 5.0 \times 10^{-5} \text{ J}$ $F = \frac{P}{v} = \frac{5.0 \times 10^{-5}}{1.0 \times 10^{-2}}$ $= 5.0 \times 10^{-3} \text{ N}$	C1 C1 A1 C1 C1 A1
8ciii	 B1 – Correct shape for B (magnitude of F around 4.0 times higher than A) B1 – Correct shape for C (magnitude of F same as A)	

Detailed explanation

From $t = 1$ s to $t = 3$ s, only one side of the frame (XY) experiences a force to the left as the induced current in XY is perpendicular to the magnetic field.

From $t = 3$ s to $t = 4$ s, the entire frame is inside region A. The frame experiences no change in magnetic flux linkage, so no emf is induced. Hence no induced current and magnetic force experienced.

From $t = 4$ s to $t = 6$ s, there is induced emf of equal magnitude but opposite polarity at both WZ and XY. Hence the magnitude of induced emf doubles to 4.0×10^{-4} V. Both WZ and XY experience a force of twice the magnitude to the left. Therefore the magnitude of F increases by 4 times (2.0×10^{-2} N).

From $t = 6$ s to $t = 7$ s, the entire frame is inside region B. The frame experiences no change in magnetic flux linkage, so no emf is induced.

From $t = 7$ s to $t = 9$ s, only one side of the frame (WZ) experiences a force to the left as the induced current in XY is perpendicular to the magnetic field. The magnitude of emf induced = $\frac{\Delta\Phi}{t} = \frac{B\Delta A}{t} = \frac{Bl\Delta s}{t} = Blv$, which is the same magnitude of emf induced from $t = 1$ s to $t = 3$ s, since v and magnitude of B are unchanged.